# Machine-checked degree lower bounds for interpolating the elliptic curve discrete logarithm from $x$-coordinates

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3 November 2026 — version 3 (draft, not human-verified; internal, not submitted)

**Abstract**

We record three degree lower bounds for polynomials that interpolate functions of the discrete logarithm from the $x$-coordinates of multiples $kP$ of a point on a short Weierstrass curve, and we give Lean 4 formalizations of them. (B) If $g(x(kP)) = k^2$ for $1 \le k \le (r-1)/2$, where $P$ has odd order $r$ on a short Weierstrass curve over a field $F$ with $2 \neq 0$ and $3 \neq 0$, then $r \le 6 \deg g + 5$. (C) If $g(x(kP)) = k$ for $1 \le k \le N$ and $2N < r$, then $N \le 3 \deg g + 2$. (D) If the interpolation $g(x(kP)) = k^2$ is required only on a set $A \subseteq [1, N]$, $2N < r$, then for each $j \ge 1$ at most $3 \deg g$ elements $k \in A$ have $k-j, k+j \in A$ (under an explicit extra hypothesis). All three are proved by one device, a cleared second-difference polynomial of exact degree $3 \deg g$ (constant $g$ is treated separately). The statements are verified by Lean against a pinned Mathlib; the Lean statements are quoted verbatim. Interpolation degree bounds of this kind are due to Coppersmith–Shparlinski, Lange–Winterhof and Satoh; we could not access their exact statements, so our bounds may be known or weaker. The contribution claimed is the machine-checked formalization and the precise hypotheses. The results concern polynomial degree in a restricted model and have no bit-complexity consequence. A clearly marked later part (Section 7) records, with their evidence levels kept separate, six further items: three formally verified barriers for restricted algebraic models (division-free straight-line programs; straight-line programs with division; computation trees with equality tests, Theorem DT, stated with its scope), a method barrier (a proof attempt, not reviewed), an experimental conjecture, and a proof-reviewed but *not formalized* bound (Theorem E). Version 3 adds a quantified *obstruction map* (Section 8) of the algorithm routes the laboratory tested in rounds 11–19 at toy scale (primes up to roughly 44 bits, with a few runs at $r \le 2^{36}$): generic search and automated discovery; index calculus for $m = 2, 3, 4$ (supply floors and finder costs); ambient groups, base change and isogenies; excess-closure factor bases; lifted relations; precomputation; integer structure, SAT and machine learning; the Mahalanobis–Mallick Problem L; and the transfer of an ETH-refutation technique. For each family it states the mechanism, the measured or derived cost copied from the ledger, the exact obstacle and the evidence level; all of it is experiment or unreviewed derivation, except the narrower E0/E3 remarks reviewed in REV-6c73a098 (decision revise). It also records a methodological lesson: naive log-log exponents at small $r$ are biased low by setup cost, so $\alpha\sqrt{r} + \gamma$ fits or a same-implementation rho control are required. **No improvement over Pollard rho is claimed anywhere in this report.** The contribution is the quantified obstruction map at toy scale together with the formal barriers for restricted models already present in version 2 (Lean-verified degree and straight-line-program bounds).

**Evidence levels used in this report.** We keep four levels apart and name the level of every item. *Formally verified (Lean)*: a Lean 4 declaration checked against the pinned Mathlib, with the statement quoted. *Reviewed proof*: a paper proof that an independent AI reviewer examined, not

formalized. *Experiment*: a measurement at toy sizes; it proves nothing about large parameters. *Conjecture*: an unproved statement. Review by AI agents of the same provider is only partially independent; no human has verified anything here.

# 1 Introduction

The elliptic curve discrete logarithm problem (ECDLP) asks, given $P$ and $Q = kP$ in a group of order $r$, for $k$ modulo $r$. A natural question about restricted classes of algorithms is whether the map $x(kP) \mapsto k$ (or a quantity such as $k^2$, which is the natural $x$-only invariant because $x(kP) = x(-kP)$) can be given by a low-degree polynomial. Here $k$ is a positive integer representative with $k \le (r-1)/2$, and the target $k^2$ (or $k$) is its image in the field $F$; it is not a function of $k$ modulo $r$ in general (over $\mathbb{F}_p$ with $p \ge 5$ and $k \le (r-1)/2 < p$ no wrap-around occurs, but over extension fields $k$ may exceed the characteristic). A lower bound on the degree of any interpolating polynomial shows that no *low-degree-polynomial formula* computes the logarithm; it is a barrier for one restricted model, in the spirit of the interpolation results cited in Section 6.

What such a bound does *not* say should be stated plainly. It says nothing about algorithms that use other operations (comparisons, branching, group operations, index calculus), nothing about lower bounds on the cost of evaluating it beyond the logarithmic one noted next, and nothing about bit complexity. Degree and evaluation cost are related only weakly: Horner's rule evaluates a degree-$d$ polynomial with at most $d$ multiplications, so degree is an upper bound for cost in that model; conversely a division-free straight-line program with $s$ multiplications computes a polynomial of degree at most $2^s$, so a degree lower bound yields only a *logarithmic* lower bound on the number of multiplications (see Section 7 for the formalized form), and no statement about bit operations. A high-degree polynomial such as $X^{p-2}$ can be cheap to evaluate. The open target of the laboratory (a universal classical ECDLP algorithm with worst-case $O(n \log n)$ bit operations) is unaffected by this report.

**Contents.** Section 2 fixes notation; Section 3 states the three results; Section 4 gives the proofs; Section 5 quotes the Lean statements and the trust base; Section 6 discusses prior work and novelty; Section 7 is a marked part with six later laboratory items (i)–(vi) (it contains the division-SLP barrier, Theorem DT and Theorem E); Section 8 gives the obstruction map of the laboratory's tested algorithm routes (rounds 11–19); Section 9 lists limitations and Section 10 concludes.

Existence is not at issue: the nodes $x(kP)$, $1 \le k \le N$, $2N < r$, are pairwise distinct (Section 2), so Lagrange interpolation always gives $g$ with $\deg g \le N-1$ for the prescribed values; for Theorem B this is $\deg g \le (r-3)/2$. Theorems B and C therefore determine the minimal interpolation degree on a full range up to a constant factor (about 3), and the bounds are linear in $r$. Theorem D is a partial-agreement count, not a minimal-degree statement on a range, and is not covered by this sentence.

# 2 Setting and notation

Let $F$ be a field and $W : y^2 = x^3 + ax + b$ a short Weierstrass cubic over $F$ ($a_1 = a_2 = a_3 = 0$, $a = a_4$, $b = a_6$). The principal case is $F = \mathbb{F}_p$ and $W$ an elliptic curve with a point $P$ of prime order $r$. For a nonzero affine point $Q$ we write $x(Q)$ for its $x$-coordinate, and we write $r = \mathrm{ord}(P)$ for the additive order of $P$. For a polynomial $g \in F[X]$ we write $m = \deg g$ and $\mathrm{lc}(g)$ for its leading coefficient. Throughout we assume $2 \ne 0$ in $F$ (written $2 \ne 0$; Theorem B also uses $3 \ne 0$).

Two facts are used. First, if $2N < r$ then $k \mapsto x(kP)$ is injective on $\{1, \dots, N\}$: $x(kP) = x(lP)$ forces $kP = \pm lP$, i.e. $k \equiv \pm l \pmod r$, and $2 \le k + l \le 2N < r$ rules out $k \equiv -l$, so $k \equiv l$,

hence $k = l$. Second, the $x$-addition identities: for $Q = (u, v)$, $P = (x_0, y_0)$ affine points with $u \neq x_0$,

$$\big(x(Q+P) + x(Q-P)\big)(u - x_0)^2 = 2\big((x_0u + a)(u + x_0) + 2b\big), \tag{1}$$

$$x(Q+P)\,x(Q-P)\,(u - x_0)^2 = (x_0u - a)^2 - 4b(u + x_0). \tag{2}$$

For (1) write $s_\pm = x(Q \pm P) = \big((v \mp y_0)/(u - x_0)\big)^2 - (u + x_0)$ and use $v^2 + y_0^2 = u^3 + x_0^3 + a(u + x_0) + 2b$ together with $(u + x_0)(u - x_0)^2 = u^3 - u^2x_0 - ux_0^2 + x_0^3$; (2) is a similar computation (it is part of the formalized algebraic core, Section 5).

## 3 Main results

**Theorem B** ($k^2$-interpolation)**.** *Let $W$ be a short Weierstrass curve over a field $F$ with $2 \neq 0$ and $3 \neq 0$ in $F$, and $P$ a point of odd order $r$. If $g \in F[X]$ satisfies $g(x(kP)) = k^2$ for $1 \leq k \leq (r-1)/2$, then $r \leq 6 \deg g + 5$; that is, $\deg g \geq (r-5)/6$.*

For an elliptic curve over $\mathbb{F}_p$ and $P$ of prime order $r > 2$ this is the statement recorded as HYP-eda6732a (the ledger statement assumes $r$ prime; the blind review REV-83ec2fc9 notes that "$r$ prime" can be weakened to "$r$ odd", which is the formalized form). The hypothesis $3 \neq 0$ is formalized but, according to REV-15104e72, not needed mathematically (see Section 4).

**Theorem C** (linear interpolation)**.** *Let $W$ be short Weierstrass over $F$ with $2 \neq 0$, $P$ a point with $\mathrm{ord}(P) = r$, and $2N < r$. If $g \in F[X]$ satisfies $g(x(kP)) = k$ in $F$ for $1 \leq k \leq N$, then $N \leq 3 \deg g + 2$. In particular, for $N = (r-1)/2$ ($r$ odd), $r \leq 6 \deg g + 5$.*

(HYP-42f91290.) The assumption $3 \neq 0$ is not needed.

**Theorem D** (partial agreement)**.** *Let $W$ be short Weierstrass over $F$ with $2 \neq 0$, $\mathrm{ord}(P) = r$, $2N < r$ (for $r$ odd, equivalently $N \leq (r-1)/2$), $A \subseteq \{1, \ldots, N\}$, and $g \in F[X]$ with $g(x(kP)) = k^2$ for all $k \in A$. For $j \geq 1$ let $S_j = \#\{k \in A : k - j \in A,\ k + j \in A\}$ (the $k$ with $j < k$ automatically). If $\deg g \geq 1$ or $j \neq 0$ in $F$, then $S_j \leq 3 \deg g$.*

(HYP-a1dd5501, part (2); the main inequality only.) The side condition is needed over general fields. An instance found by the adversarial review REV-15104e72 (checked there by computer, not by us and not in Lean): over $\mathbb{F}_{27} = \mathbb{F}_3[t]/(t^3 - t + 1)$ the curve $y^2 = x^3 + t^2x$ has a point $P$ of order 28; with $N = 7$ (so $2N = 14 < 28$), $A = \{1, 4, 7\}$, $g = 1$ and $j = 3$ one has $g(x(kP)) = k^2$ on $A$ ($k^2 \in \{1, 16, 49\} = \{1\}$ in characteristic 3), $\deg g = 0$ and $S_3 = 1 > 0$. This instance needs $r > 4p + 2$, which is impossible over $\mathbb{F}_p$ by the Hasse argument below; that argument does not extend to $\mathbb{F}_q$. Over $\mathbb{F}_p$ the condition excludes only constant $g$ with $p \mid j$, and then $S_j = 0$ for the following reason (*not* formalized in Lean): $S_j \geq 1$ requires $k - j \geq 1$ and $k + j \leq N$, so $N \geq 2j + 1 \geq 2p + 1$, whence $r > 2N \geq 4p + 2$, contradicting the Hasse bound $r \leq \#E(\mathbb{F}_p) \leq p + 1 + 2\sqrt{p}$. Hence over $\mathbb{F}_p$ the unconditional informal statement $S_j \leq 3 \deg g$ holds. (One must not argue $j \leq N < p$; that is false in general, e.g. $y^2 = x^3 + 2x + 1$ over $\mathbb{F}_3$ with $r = 7$, $N = 3$.)

*Remark* 1 (Hypotheses and the refuted transcription). The hypothesis $2N < r$ in Theorem D cannot be dropped. The ledger record HYP-5da6025e (status: refuted) contains a transcription of part (2) with $N < \min(p, r)$ in place of $N \leq (r-1)/2$; the blind review REV-ac47cc67 produced the counterexample $y^2 = x^3 + 5$ over $\mathbb{F}_7$, $P = (3, 2)$ of order $7 = p$, $A = \{1, \ldots, 6\}$, $g = 5X$, which satisfies $g(x(kP)) = k^2$ for $k = 1, \ldots, 6$ with $\deg g = 1 < S_1/3 = 4/3$. The hypothesis $N \leq (r-1)/2$ had been present in the original agent report and was lost in a transcription error by the principal investigator; the corrected statement is HYP-a1dd5501 and is what is formalized here. Ledger parts of HYP-a1dd5501 not covered here: part (1) for rational functions on $E$ (reviewed, not formalized), and the density corollary $\deg g \geq ((3\delta - 2)N - 2)/3$ for $\delta > 2/3$ (reviewed, not formalized).

*Remark* 2 (Background: Theorem A). The ledger also records (HYP-adb581ff, status *open*, reviewed, not formalized) a result for rational functions $f$ on $E$ with $f(kP) = k$ for $1 \le k \le N \le r-1$: the pole-divisor degree is at least $(N-1)/2$, and polynomial total degree at least $(N-1)/6$. It is recorded as known in kind (Section 6) and is cited only as background.

# 4 Proofs

Assume $m = \deg g \ge 1$ for the moment; constant $g$ is not covered by the exact-degree argument and is treated separately at the end of each case. Fix an affine point $P_0 = (x_0, y_0)$ ($P_0 = P$ in Theorems B, C; $P_0 = jP$ in Theorem D) and put $d = u - x_0$.

**The cleared second-difference polynomial.** For a variable $u$ let $s_\pm$ be the (formal) roots of $z^2 - e_1 z + e_2$ where $e_1 = s_+ + s_-$ and $e_2 = s_+ s_-$ are the rational functions in $u$ given by (1) and (2) (so $g(s_+) + g(s_-)$ is *defined* as the symmetric expression $\sum_i c_i p_i(e_1, e_2)$ below, and is evaluated at $u = x(Q)$ only after clearing denominators): $e_1 = n_1/d^2$, $e_2 = n_2/d^2$ with $n_1, n_2$ polynomials of degree $\le 2$. Write $g = \sum_{i \le m} c_i X^i$. Then $g(s_+) + g(s_-) = \sum_i c_i p_i$ with $p_i = s_+^i + s_-^i$ a polynomial with integer coefficients in $e_1, e_2$ whose monomials $e_1^\alpha e_2^\beta$ satisfy $\alpha + 2\beta = i$ (Newton–Girard). Multiplying a monomial by $d^{2m}$ gives $n_1^\alpha n_2^\beta d^{2(m-\alpha-\beta)}$, a polynomial of degree $\le 2(\alpha+\beta) + 2(m-\alpha-\beta) = 2m$, because $\alpha + \beta \le i \le m$. Define

$$G_c(u) = d^{2m}\big[g(s_+) + g(s_-)\big] - 2d^{2m} g(u) - c\, d^{2m} \in F[u],$$

for a constant $c$. The first term has degree $\le 2m$, the third degree $\le 2m$ (and 0 when $c = 0$), and $-2d^{2m} g(u)$ has degree exactly $3m$ with leading coefficient $-2\,\mathrm{lc}(g) \ne 0$ (here $2 \ne 0$ is used). Since $2m < 3m$ for $m \ge 1$:

$$\deg G_c = 3m \quad \text{for every constant } c, \quad \text{in particular } G_c \ne 0. \tag{3}$$

**Roots.** Suppose $Q = kP$ is affine with $x(Q) \ne x_0$ and $Q \pm P_0$ affine. Then evaluating at $u = x(Q)$, the quantities $s_\pm$ are $x(Q \pm P_0)$ by the identities of Section 2, so $G_c(x(Q)) = d^{2m}\,[g(x(Q+P_0)) + g(x(Q-P_0)) - 2g(x(Q)) - c]$ with $d \ne 0$.

*Theorem B.* Take $P_0 = P$, $c = 2$. For $2 \le k \le (r-3)/2$ we have $1 \le k-1$ and $k+1 \le (r-1)/2$, so all three values of $g$ are prescribed and the bracket equals $(k+1)^2 + (k-1)^2 - 2k^2 - 2 = 0$. The values $x(kP)$, $2 \le k \le (r-3)/2$, are pairwise distinct (injectivity) and differ from $x(P)$ (since $k \ge 2$), so $G_2$ has at least $(r-5)/2$ distinct roots. By (3), $(r-5)/2 \le 3m$, i.e. $r \le 6m+5$. Constant $g = c_0$: if $r \le 5$ the bound is trivial; if $r \ge 7$ then $(r-1)/2 \ge 3$ and $g(x(P)) = 1$, $g(x(2P)) = 4$ force $3 = 0$, which $3 \ne 0$ excludes. (According to REV-15104e72, $3 \ne 0$ is superfluous: in characteristic 3 and $r \ge 7$, $k = 3$ gives $9 = 0 \ne 1 = g$. We keep it because it is the formalized hypothesis.) Hence the bound holds.

*Theorem C.* Take $P_0 = P$, $c = 0$. For $2 \le k \le N-1$ the bracket is $(k+1) + (k-1) - 2k = 0$, giving $N-2$ distinct roots (injectivity as $2N < r$), so $N - 2 \le 3m$. If $g$ is constant and $N \ge 2$, then $g(x(P)) = 1 \ne 2 = g(x(2P))$, contradiction; so $N \le 1 \le 3m + 2$.

*Theorem D.* Take $P_0 = jP$ (note $j < k \le N < r$). For $k \in C_j = \{k \in A : j < k,\ k \pm j \in A\}$ the bracket with $c = 2j^2$ is $(k+j)^2 + (k-j)^2 - 2k^2 - 2j^2 = 0$. The values $x(kP)$ are distinct for distinct $k \in C_j$ and differ from $x(jP)$ because $k \ne j$ and $k + j \le 2N < r$, so $|C_j| \le 3m$ by (3). If $g = c_0$ is constant, the bracket is $-2j^2 \ne 0$ whenever $j \ne 0$ in $F$, so $G$ has no roots and $C_j = \emptyset$. □

**Exact degree.** The computation above shows that, for the second-difference identity, the cleared polynomial has degree exactly $3 \deg g$ and leading coefficient $-2\,\mathrm{lc}(g)$; for Theorem C this is recorded in HYP-42f91290 and agrees with the numerical checks of the ledger.

## 5 Formal verification

**Environment.** Toolchain `leanprover/lean4:v4.35.0` (file `lean-toolchain`); Mathlib revision `56a005855ea17e0b0958f73ec801ae140b0a1679` (`lakefile.toml`, with `-DwarningAsError=true`). Source commits recorded in the proof records: Theorem B, PRF-34538f75, commit `cc8dc791a0bd74f89b6cd614898a5773abdc500a` (`lake build`: 3163 jobs, 0 warnings); Theorem C, PRF-35d1387a, commit `8191c0f69398b632fc2c7c7efbfdcdb9e2634b7b` (3164 jobs, 0 warnings); Theorem D, PRF-5357b698, commit `84ddbad82b2f1412e851cf1ab6012b6650bba544` (3165 jobs, 0 warnings). Files: `MathResearch/ECDLP/Interpolation.lean` (generic polynomial core), `CurveX.lean`, `LinearInterp.lean`, `PartialAgreement.lean`. All three proofs use Mathlib's `WeierstrassCurve.Affine.Point`; the curve-specific input is the $x$-addition identities (`CurveX.sum_prod_identity`, `CurveX.smul_relations`) and injectivity of $k \mapsto x(kP)$.

**Statements (verbatim from the repository).** Here `xCoord` is the $x$-coordinate function on `W.Point`, defined to be 0 at the point at infinity (harmless here, since every point evaluated is $kP$ with $1 \le k < r$, which is affine). The binders `{F : Type*} [Field F] [DecidableEq F]` come from `variable` lines of the files; `Affine` is `WeierstrassCurve.Affine` (via `open WeierstrassCurve`) and `F[X]` is `Polynomial F`.

The declarations live in the Lean namespace `MathResearch.ECDLP.CurveX`, in these files: Theorem B in `CurveX.lean`, Theorem C in `LinearInterp.lean`, Theorem D in `PartialAgreement.lean`.

```
theorem degree_bound_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (h3' : (3 : F) ≠ 0) (P : W.Point) (r : ℕ)
    (hr : addOrderOf P = r) (hodd : r % 2 = 1) (g : F[X])
    (hg : ∀ k : ℕ, 1 ≤ k → k ≤ (r - 1) / 2 → g.eval (xCoord (k • P)) = (k : F) ^ 2)
     :
    r ≤ 6 * g.natDegree + 5
```

```
theorem degree_bound_linear_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂
     = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ)
    (hr : addOrderOf P = r) (N : ℕ) (hN : 2 * N < r) (g : Polynomial F)
    (hg : ∀ k : ℕ, 1 ≤ k → k ≤ N → g.eval (xCoord (k • P)) = (k : F)) :
    N ≤ 3 * g.natDegree + 2
```

```
theorem partial_agreement_degree_bound {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (A : Finset ℕ) (hA : ∀ k ∈ A, 1 ≤ k ∧ k ≤ N) (g : F[X])
    (hg : ∀ k ∈ A, g.eval (xCoord (k • P)) = (k : F) ^ 2) (j : ℕ) (hj : 1 ≤ j)
    (hjg : 1 ≤ g.natDegree ∨ (j : F) ≠ 0) :
    (A.filter (fun k => j < k ∧ k - j ∈ A ∧ k + j ∈ A)).card ≤ 3 * g.natDegree
```

The full names are `MathResearch.ECDLP.CurveX.degree_bound_short_weierstrass`, `MathResearch.ECDLP.CurveX.degree_bound_linear_short_weierstrass` and `MathResearch.ECDLP.CurveX.partial_agreement_degree_bound`.

**Correspondence to the informal statements.** (i) The formal statements are for an arbitrary field $F$, not only $\mathbb{F}_p$, and for any point $P$ of the given additive order; $r$ need not be prime. Theorem B needs $r$ odd, $2 \neq 0$ and $3 \neq 0$; Theorems C and D need neither $3 \neq 0$ nor $r$ odd (the condition $2N < r$ is equivalent to $N \le (r-1)/2$ for $r$ odd). (ii) Since `W : Affine F` is not assumed nonsingular, the review REV-1f44f52b notes that the results also cover nonsingular points of singular cubics. (iii) Theorem D carries the extra hypothesis `hjg`; the Hasse-bound

argument that removes it over $\mathbb{F}_p$ (Section 3) is *not* formalized. (iv) The formal statements are degree bounds only; they assert no existence of $g$ and no complexity. REV-1f44f52b and REV-6c9bfbc1 accepted the correspondence of B and C; the independent review REV-685408b5 (formal correctness only; no complexity or novelty verification) verified the correspondence for D.

**Trust base.** `#print axioms` on each of the three declarations reports exactly `[propext, Classical.choice, Quot.sound]`. The evidence is per theorem (we verified that each file contains the corresponding `#print axioms` line): Theorem B, `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-b585f279-fd16-4cca-b3a1-c2484d545078.evidence` (`degree_bound_short_weierstrass`, together with `sum_prod_identity` and `smul_relations`); Theorem C, `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-5da55f17-bbac-4f82-9271-da5f30d66b80.evidence` (`degree_bound_linear_short_weierstrass`); Theorem D, `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-ca0a1a26-2ee3-4da0-be6c-96a56ffa6cc6.evidence` (`partial_agreement_degree_bound`). Trust therefore rests on the Lean kernel, Mathlib at the pinned revision, and the faithfulness of the statements to the informal ones (reviewed by AI agents of the same provider; independence is partial).

**Reproduction.** At a checkout of the repository: `lake build` (pinned toolchain and Mathlib), followed by `#print axioms MathResearch.ECDLP.CurveX.degree_bound_short_weierstrass` and likewise for the other two names (see `research/ecdlp/proofs/checks/Round8Axioms.lean` for the pattern).

**Empirical sanity checks (not part of the proof).** The ledger records toy-size numerical checks: the $x$-addition identity 2519 times with no failure (Theorem B record); 208 toy cases without violation (REV-83ec2fc9); for Theorem C 533 curves with $p \leq 211$, then 12,459 cases including extension fields (REV-6c9bfbc1); 1242 toy points for B (REV-1f44f52b); 739,847 pairs $(A, j)$ for D (a weak check). These tests do not constitute proofs and are not needed given the Lean proofs; they indicate the bounds are far from tight in tests.

## 6 Relation to prior work and novelty status

Coppersmith and Shparlinski [1] studied polynomial approximation of the discrete logarithm and the Diffie–Hellman mapping (title of [1]); we did not access the text. Lange and Winterhof (COCOON 2002 [6]; AAECC-15 2003 [7]) and Satoh (WCC 2005, published 2006 [11]) prove degree lower bounds for polynomial interpolation of the elliptic curve discrete logarithm and related maps. **The exact statements of these results were not accessible to the laboratory**: only bibliographic records and abstracts were seen, and full texts were not accessed. We therefore cannot compare constants or hypotheses. Our bounds may be known or weaker. Theorem A is recorded in the ledger as known in kind; the laboratory's novelty checks (ECDLP-22) found no result of the form of Theorems B–D in the accessible material, but this is absence of evidence, not a novelty claim. The laboratory's ledger record for Theorem C (HYP-42f91290) says it is "likely implied by / weaker than Lange–Winterhof-type results" and is *not* claimed novel. Novelty of all three is *unsettled*.

**Claimed contribution:** a machine-checked formalization (Lean 4, Mathlib, standard axioms only) of the three degree bounds, with explicit, reviewed hypotheses (including the corrected hypothesis $2N < r$ of Theorem D and its generality over arbitrary fields). Bibliographic metadata (first names, venue, pages, year, DOI) of the cited entries were checked against Crossref records (by REV-2b383b9b for [6], and in this revision for [7], [11] and [1]); in particular the year of [11] is 2006. Full texts were still not accessed.

# 7 Later laboratory results (not part of Theorems B–D)

This section was added in revision 1 and extended in revision 2. It is clearly separate from the results above. Its six items (i)–(vi) carry the statuses stated in the headings; the statuses are those recorded in the laboratory ledger and are not upgraded here.

**(i) Division-free straight-line programs: formally verified; independently reviewed for formal correctness.** Ledger record PRF-58cf340e (derived from HYP-92a9b4d9; status *formally verified*; the independent batched review REV-7f70c161 (decision *accept*, formal correctness only) targets it. Revision 1 of this report said this review was pending; the ledger now records it, so the statement is updated here). In `MathResearch/ECDLP/SLPBound.lean` a *program* over a commutative ring $F$ is a list of operations with a single input variable $x$; each operation appends an arbitrary constant of $F$, the input, or the sum, difference or product of two earlier values; every operation, including `const` and `var`, counts one step toward the length $L$ (the Lean docstring calls constants "free" only in the sense that any element of $F$ may appear; the `const` operation itself is counted, as REV-7f70c161 also records). There are no divisions, comparisons or branching. A program of length $L$ computes a polynomial of degree at most $2^L$ in $x$. Combined with Theorem C this gives, in Lean (source commit `bff5e14c0f187719fb79c07f7768a2856f20604a`, `lake build` 3166 jobs, 0 warnings; build evidence `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-94bd6bd6-9c44-47e7-9743-cd567e803d9f.evidence`; axioms `[propext, Classical.choice, Quot.sound]` in `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-aeddfabd-5935-4ad5-9911-2c48a87063dc.evidence`):

```
theorem length_bound_linear_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂
    = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (prog : List (Op F)) (i : ℕ)
    (hprog : ∀ k : ℕ, 1 ≤ k → k ≤ N → (evalAt prog (xCoord (k • P))).getD i 0 = (k
    : F)) :
    N ≤ 3 * 2 ^ prog.length + 2
```

(namespace `MathResearch.ECDLP.SLPBound`; `evalAt` evaluates the program on the input, and out-of-range indices read 0.) Thus a program of length $L$ that outputs $k$ at $x(kP)$ for $1 \le k \le N$, $2N < r$, needs $L \ge \log_2((N-2)/3)$. This answers directly the distinction between polynomial degree and program length raised in review REV-15104e72 (N1): the bound is on the number of ring operations in this restricted model, logarithmic in $N$. The cost-model consequence recorded in the ledger (if each field operation costs $\Omega(\log p)$ bit operations, then $\Omega(n^2)$ bit operations are needed in this model) is a remark, *not formalized*, and concerns only this restricted model; it says nothing about the open $O(n \log n)$ target for general algorithms.

**(ii) Straight-line programs with division: formally verified; independently reviewed for formal correctness.** Ledger records HYP-6aef8c30 (status *completed*, outcome *verified*, proof status *formally verified*) and PRF-9586acb8; the independent batched review REV-7f70c161 (decision *accept*) covers formal correctness only. Source commit `3ecf1f474ae3868e877727d1061c444c5307271f`, `lake build` 3167 jobs, 0 warnings (build evidence `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-3aa1a646-4c3d-4ffa-b511-6c3b1ba10f31.evidence`); axioms `[propext, Classical.choice, Quot.sound]` (`research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-ec79161e-5786-4124-9252-7d36a4adcc4e.evidence`). In `MathResearch/ECDLP/RationalSLP.lean` the operation type `Op` additionally has a division `div i j`; values are evaluated in $F$ (with Lean's total convention $x/0 = 0$), and `Defined prog u` says that *every* division of `prog` (including dead ones) has a nonzero divisor value at $u$. The

statement, verbatim from the repository (namespace `MathResearch.ECDLP.RationalSLP`; the binders `{F : Type*} [Field F] [DecidableEq F]` come from a `variable` line):

```
theorem length_bound_rational_slp {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (prog : List (Op F)) (i : ℕ)
    (hdef : ∀ k : ℕ, 1 ≤ k → k ≤ N → Defined prog (xCoord (k • P)))
    (hprog : ∀ k : ℕ, 1 ≤ k → k ≤ N → (evalAt prog (xCoord (k • P))).getD i 0 = (k
    : F)) :
    N ≤ 3 * 2 ^ prog.length + 2
```

In words: for a straight-line program over $F$ with $+, -, \times, \div$ and constants and a single input $u$, if every division has a nonzero divisor at $u = x(kP)$ and the $i$-th output equals $k$ there for all $1 \le k \le N$, $2N < r$, then $N \le 3 \cdot 2^L + 2$ where $L$ is the program length (so $L \ge \log_2((N-2)/3)$). The constant is the same as in the division-free case (i). The proof (as recorded) obtains from the program a pair of polynomials $(A, B)$ of degree $\le 2^L$ with $A(x(kP)) = k\,B(x(kP))$ and $B(x(kP)) \ne 0$, and applies a rational analogue of Theorem C (`degree_bound_rational_short_weierstrass`: $N \le 3d + 2$ for $\max(\deg A, \deg B) \le d$, no coprimality hypothesis assumed); the non-vanishing of the cleared polynomial uses orbit propagation, which forces $2 = 0$ in $F$. The review REV-7f70c161 records that the bound $3d + 2$ is attained at $d = 1$ ($p = 11$, $y^2 = x^3 + 1$, $P = (9, 2)$, $A = 1 + u$, $B = 3 + 2u$, $N = 5$). The Lean statement for odd $r$ in the form $r \le 6 \cdot 2^L + 5$ is `length_bound_rational_slp_order` in the same file (not separately recorded in the ledger records we consulted). *Scope.* The model has no branching or comparison, no bit access, and does not take $P$ as input; it counts field operations only and makes no bit-complexity claim. It does not apply to rho, baby-step giant-step or index calculus. The ledger's earlier remark that the true constant might be nearer 2 is refuted as a uniform claim by the attained instance above.

**(iii) Theorem DT: algebraic computation trees with equality tests; formally verified, paper proof reviewed (the correspondence of the Lean statement to the paper statement has not been independently reviewed).** Ledger record HYP-dbe79a50 (status *open*, proof status *formally verified*, outcome *verified*), Lean record PRF-ab954d65; the blind adversarial review REV-d7bc5940 confirmed the paper proof of Theorem DT and independently found the gap in the paper proof of its Lemma A that the Lean proof repairs (propagation from one adjacent pair in both directions). Source commit `b699c3e8ac3ba3d96b1f6b6bc0fcf7b3eea34cbd`, `lake build` 3168 jobs, 0 warnings (build evidence `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-9ccbf89a-509f-496d-b41f-a352be9c57c5.evidence`); axioms `[propext, Classical.choice, Quot.sound]` (`research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-c8fc5d7e-6598-4dbb-81dc-f3bc94eeb0f2.evidence`). Lean source `MathResearch/ECDLP/DecisionTree.lean`. The Lean statement, verbatim (namespace `MathResearch.ECDLP.DecisionTree`; same binders as above):

```
theorem Tree.length_bound {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (T : Tree F) (L : ℕ) (hL : T.arithDepth ≤ L)
    (hrun : ∀ k : ℕ, 1 ≤ k → k ≤ N → T.exec (xCoord (k • P)) = some (k : F)) :
    N ≤ 9 * 2 ^ L - 1
```

The informal statement recorded in the ledger (verbatim from the HYP record): *Theorem DT (algebraic decision trees with equality tests): over a field $F$ with $2 \ne 0$, $W$ short Weierstrass, $P$ of order $r$, $2N < r$. Any algebraic computation tree with single input $u = x(Q)$, arithmetic nodes $+, -, \times, \div$ (divisions defined where executed) and branching only on "register $= 0$?", in which every root-to-leaf path has at most $L$ arithmetic nodes, and which outputs $k$ at $u = x(kP)$ for all $1 \le k \le N$, satisfies $N \le 9 \cdot 2^L - 1$ regardless of depth.* The formal model: a `Tree` is a leaf (outputting a register), an arithmetic node (an `Op`: `const`, `var`, `add`, `sub`, `mul`, `div`), or a test

"register = 0?" with two subtrees; register 0 is the input; `arithDepth` is the maximal number of arithmetic nodes on a root-to-leaf path and tests are not counted; `exec` returns `none` if an executed division has a zero divisor.

*Scope (after the complexity review REV-6c73a098 of remarks E0 and E3).* Theorem DT is a statement about *equality branching only*: the only tests are "register = 0". *Constants are charged*: in the formal model every `const` node is an arithmetic node and counts toward $L$ (the informal text above does not say this; the ledger note and the review do). With *order comparisons* the arithmetic length becomes vacuous: the review records that a binary-search tree over the sorted values has 0 arithmetic nodes (toy $N = 300$: depth 9, 300 leaves), so in a model with comparisons only tree size and depth matter, and no size–depth lower bound is proved. Theorem DT therefore does *not* bound algorithms that compare integers or inspect bits of field elements, and it bounds arithmetic nodes, not tree size, depth or running time. The remarks about Shoup's generic group model recorded with Frontier E (E0) are reviewed only as heuristics and are not used here. The ledger also records Theorem P and Corollary P1 for this model; they are *not* formalized, Corollary P1 was false as transcribed (missing hypothesis $\operatorname{char} F = 0$ or $\operatorname{char} F > N$; counterexample over $\mathbb{F}_{3^7}$) and is superseded by HYP-8e55674a; we do not state them. Novelty of Theorem DT is unchecked (the ledger notes Ben-Or/Strassen-type degree arguments and Lange–Winterhof as possibly related).

**(iv) Method barrier L2: proof attempt, PI-checked, not reviewed.** Ledger record HYP-0d4f14da (status *open*, proof status *proof attempt*; the principal investigator checked the proof steps and reproduced the supporting numerical scripts; not formalized, not independently reviewed). The claim, as recorded: for a "translation annihilator" $D = \sum_{i \in S} c_i G(Q + iP)$ with $\sum c_i = \sum i c_i = 0$ ($|S| = s$, width $w$), $G = g \circ x$, $\deg g = d \geq 1$, $2N < r$, if $D \not\equiv 0$ then $D$ has pole degree exactly $2ds$ and at least $2(N - w)$ zeros, so $N \leq ds + w$, minimized at $s = 3$, $w = 2$, which is Theorem C. The zeros arise at the points $Q = kP$ where all translates are in the prescribed range: there $D(kP) = \sum_i c_i f(k + i) = 0$ because the prescribed values $f(k) = k$ have degree $\leq 1$ and $\sum c_i = \sum i c_i = 0$ (we spell this out; the ledger states only the zero count). Hence zero-counting arguments with translates cannot improve the constant 1/3; any sharper bound must use information beyond the pole structure and the values $G(kP) = f(k)$. We state this only as recorded; it has no proof in this report. The same record contains a claim L3 on anomalous $j = 0$ curves which we do not use.

**(v) Empirical conjecture C* (not proved).** Ledger record HYP-a8d1f541 ("Conjecture C*", status *open*, experimental): outside structured families (for example anomalous CM $j = 0$ curves), the minimal degree of an interpolating $g$ is $N - 1 - O(1)$. The ledger records 154,288 toy cases over prime fields with $p \leq 997$ and a maximal observed deficit of 3. This is an observation at toy sizes, not evidence for large parameters. If true it would show that the factor 3 between the bounds above and the trivial Lagrange upper bound is a limitation of the method, not of the problem.

**(vi) Theorem E (cusp-degeneration bound): reviewed proof, *not formalized, not formally verified.*** Ledger record HYP-a4caa1de ("Theorem E (corrected after review)", status *completed*, proof status *reviewed*, outcome *verified*; it supersedes HYP-38005bec, where the same statement appeared under the label "Theorem D"). One blind review, REV-3b9427bd (decision: revise, i.e. accept with three statement corrections; the corrected statement is the one below), examined it by symbolic recomputation for $N = 3, 4$ and exhaustive small-$p$ tests; the proof is a paper proof using the Schwartz–Zippel lemma (listed in the ledger as an assumption) and the standard theory of division polynomials, taken as standard and not re-proved. No Lean formalization exists, no human has checked it, and it is *not* a formally verified result. The ledger claim, verbatim from the record:

(Theorem D, corrected) Let $N \geq 3$, $p > 2N$ prime, $V := \{\tau \in \mathbb{F}_p^3 : 4A^3 + 27B^2 \neq 0,\ W(\tau) \neq 0\}$ (the valid triples, on which $\mathrm{mindeg}_N$ is defined and equals deg of the interpolant), $Z := \{\tau \in V : c_{N-1}(\tau) = 0\}$. If $p$ does not divide $\mathrm{Num}(\tau_0)$ (always true for target $k^2$; for target $k$ iff $p \notin \mathrm{Bad}_N$), then $|Z| \leq w_N p^2$, hence $\mathrm{mindeg}(N) = N - 1$ for all $\tau \in V \setminus Z$. [$|\mathbb{F}_p^3 \setminus V|$ is separately bounded by $(6 + \deg W)p^2$ and is not covered by this theorem.]

Notation (from the review report `research/ecdlp/runs/RUN-d55e8925-6e2d-41b4-8548-10a111df3297.md`, which also gives the corrected statement): $\tau = (A, x_0, y_0)$ with $B := y_0^2 - x_0^3 - Ax_0$ (the curve is $y^2 = x^3 + Ax + B$ and the point is $(x_0, y_0)$); $\psi_m$ are the division polynomials; $W := \prod_{1 \leq a < b \leq N} \psi_{a+b}\psi_{b-a}$; $c_{N-1}(\tau)$ is the divided difference of order $N-1$ of the target function ($k^2$ or $k$) at the nodes $x(kP)$, $1 \leq k \leq N$, so that $c_{N-1}$ is the leading coefficient of the interpolant; $\mathrm{Num} := W\, c_{N-1}$ as an explicit polynomial; $\tau_0 = (A, x_0, y_0) = (0, 1, 1)$ is the cuspidal point; $\mathrm{Bad}_N := \{p > 2N \text{ prime} : p \mid \mathrm{Num}(\tau_0)\}$ (a finite set of primes, the primes $> 2N$ dividing the numerator of $c_{N-1}(0)$ for target $k$; the review lists, e.g., $\mathrm{Bad}_3 = \{11\}$ and $\mathrm{Bad}_4 = \{151\}$); $w_N$ is half the weighted degree of Num (a bound for its total degree after substituting $B$). The review's values are $w_3 = 23$ and $w_4 = 81$. *Scope.* The theorem concerns fixed small $N$: the bound $w_N p^2$ is non-vacuous only for $p$ large compared with $N$ (the review notes $N \lesssim (3p)^{1/4}$ for non-vacuity); in the exhaustive tests it was genuinely tested only for $N = 3$, $p \in \{29, \ldots, 47\}$. The ledger records that it does not prove Conjecture C* beyond the regime $N < p^{1/4}$, that it is not an algorithm, and that it gives no complexity claim.

# 8 Obstruction map for algorithm candidates (rounds 11–19)

This section is *not* a theorem section. It maps every algorithmic route the laboratory tested in rounds 11–19 to the obstacle that stopped it. Each ledger hypothesis (HYP) record summarized here has recorded outcome *refuted* (the candidate was tested and gave no asymptotic improvement over Pollard rho, at most constant-factor or amortized gains of known techniques, or the tested signal was null), except the literature-survey record HYP-7c98bfbd, whose outcome is *verified* and proof status *not-applicable*. All numbers are copied in substance from the HYP records named in each paragraph; we have *not* recomputed them. "rho" is Pollard rho on the same toy instances unless stated. The map is organized by *mechanism family*. For each family we give the mechanism, the cost measured or derived, the exact obstacle, and an **evidence level** from the four used in this report: *experiment* (toy-size measurement), *unreviewed derivation* (a paper argument no reviewer has examined), *reviewed proof* or *formal proof*. **No family reaches the last two levels**, with one marked exception (the reviewed E0/E3 remarks, Section 7(iii)); every experimental item is a toy Python implementation with no independent review record (the ledger notes of the records say so explicitly). None of these results proves that no better algorithm exists, and none establishes a property for large parameters. **No asymptotic improvement over Pollard rho is claimed, and none was found.**

**Literature citations and read levels.** Literature is cited only where the ledger records it as retrieved (survey record HYP-7c98bfbd, source table in `research/ecdlp/experiments/round17-literature/survey.md`, retrieved 2026-11-03). Read levels there are: **FT** full text obtained (cited passages read, not necessarily every page), **AB** abstract or landing page only (a second-hand summary produced by a small model), **SN** search snippet only. Each bibliography entry states its level. Sources recorded only as SN or as not retrieved are not cited. Bibliographic metadata of Section 8 entries come from the survey table and were *not* checked against Crossref.

## 8.1 Family 1: generic search, rho variants and automated discovery (frontiers G and H)

**Mechanism.** Frontier G (HYP-a73b4fdb, ECDLP-54): evolutionary search over a 19-operation domain-specific language (group operations, comparisons and bits of $x$, a bounded hash ta-

| Family | Ledger records | Evidence level | Outcome at toy scale |
|---|---|---|---|
| 1 Generic, rho variants, automated discovery | HYP-a73b4fdb (frontier G), HYP-886dfa70 (frontier H) | experiment | rediscovers rho/BSGS; no exponent below 1/2 |
| 2 Index calculus, $m = 2, 3, 4$ | HYP-720e359e (D), HYP-3c665e7b (E), HYP-be407930 (F), HYP-4825a5fb (I), HYP-7c98bfbd (survey) | experiment; unreviewed derivations | closed for $m = 2, 3, 4$ |
| 3 Ambient groups, base change, isogenies | HYP-ae6977f0 (frontier F), HYP-d6cc9bce (B), HYP-8b156dfd part 2 (frontier I) | experiment; unreviewed lemma | constant factors only |
| 4 Excess-closure factor bases | HYP-73c24d12 (G) | experiment; unreviewed derivation | excess closure carries no rank or needs logs |
| 5 Lifted relations | HYP-23351e75 (A) | experiment | slower than rho at 24 bits |
| 6 Precomputation | HYP-d913fa61 (C), HYP-bfb7d000 (H) | experiment | stays on $ST^2 \approx r$ |
| 7 Integer structure, SAT, ML | HYP-6da71353 (E), HYP-93fbd7d0 (D) | experiment; E0/E3 reviewed (narrower form) | null |
| 8 Problem L | HYP-8b156dfd part 1 (frontier I) | experiment | zero-sum subset search, $\sim r$ |
| 9 ETH-technique transfer | HYP-32e353ac | unreviewed derivations | negative on every route |

Table 1: Overview of the obstruction map. The first column of each row is the family; the details and all numbers are in the corresponding paragraph.

ble, counters), trained on 10–12-bit curves and tested on held-out sizes 13–16 and fresh curves. Frontier H (HYP-886dfa70, ECDLP-57): the same idea with field arithmetic, $S_3$, lifting and linear-algebra instructions and a corrected fitness (pre-registered rule T1–T4, trained at $2^{10}$–$2^{12}$, fitted $2^{13}$–$2^{16}$, held out $2^{17}$–$2^{20}$, 3 curves × 40 instances, no log-log exponents).

**Cost.** G: nothing beats exponent 1/2 once fixed setup cost is modelled. The evolved champion s1 (a rho-like table-stepped walk keyed by low bits of $x$ with hash collision detection, success 1.00) literally passed the pre-registered rule (test-A $\beta = 0.375$ $[0.326, 0.426]$ below the BSGS lower 99% bound 0.434), but hand distinguished-point rho in the same language scores 0.334 $[0.295, 0.372]$, and on a fresh curve per size both fit cost $\sim 1.77\sqrt{r} + 136$ (PI refit); s3 has $\beta = 0.456$ $[0.391, 0.519]$; s0, s2 are not solvers; no program rediscovered BSGS. H: the rho control is $1.80\sqrt{r}+130$ (reproducing round 16); all four evolved champions fail (r0 rho-with-memory: ratio to the control 1.16 to 2.0, success 0.82 at $2^{20}$; r1 the empty program; s2 a BSGS variant: ratio $\sim$ 0.65–0.7, flat, slope 0.014 $[-0.041, 0.075]$; s3 a BSGS variant: success 0.83 at $2^{18}$); no champion uses field, $S_3$, lifting or linear-algebra instructions with any effect (dead code). A hand-written index-calculus program in the same language solves instances but its cost ratio to rho grows from about 34 to 96 over $2^{10}$–$2^{14}$ (cost/$\sqrt{r}$ 127 to 266 at 11–14 bits versus 4.5 to 2.8 for rho; success falls 0.95 to 0.73) and it is selected out of seeded populations.

**Obstacle.** The only apparent sub-1/2 exponent in G is setup amortisation, not a faster algorithm; in H the expressible index calculus loses by a growing factor and the search never selects field instructions.

**Evidence level.** Experiment, closed for these languages, search budgets (H: 4 runs × 120 generations) and sizes only. Ledger caveats: in G only random seeds 0–3 completed and the supplement stops at $2^{16}$ (both runs stopped by a 270 s timeout); in H the 99% interval at $2^{20}$ is wide (BSGS $Q = 0.96$ $[0.79, 1.16]$), and bug fixes to `evaluate2.py` were made after protocol hashing (criteria unchanged). No independent review.

**Methodological note: naive log-log exponents at small $r$ are biased low (frontier G).** Frontier G's lesson, recorded by the PI in HYP-a73b4fdb: naive log-log exponent fits at $r \le 2^{16}$ are biased low by additive setup cost; hand rho gives 0.327 and BSGS gives 0.438 even though both are $\Theta(\sqrt{r})$. The champion's pre-registered "pass" ($0.375 < 0.434$) was exactly this artefact: its *same-implementation* control, hand rho, scores lower still, and a model $\alpha\sqrt{r} + \gamma$ fits both with identical $1.77\sqrt{r}+136$ (BSGS: $1.51\sqrt{r}+33$). *Any speedup claim must therefore fit $\alpha\sqrt{r}+\gamma$, or compare with a same-implementation rho control, before reporting an exponent.* Frontier H's protocol adopted this rule. We flag that several other fitted exponents quoted below (inventor B1 in Family 3, inventor C and the other fitted exponents in Family 6) are naive log-log or local fits at toy sizes; the ledger records some with confidence intervals but does not say that setup cost was modelled, so they should be read as upper-level indications only, not as asymptotic exponents. We have not re-fitted them.

### 8.2 Family 2: index calculus for $m = 2, 3, 4$ (inventors D, E, F, I; literature survey; PI derivations)

**Mechanism.** Decompose a point as a sum of $m$ points of a factor base $V$ of size $s$ using Semaev summation polynomials ($S_3$ for $m = 2$, $S_4$ for $m = 3$, $S_5$ for $m = 4$), collect relations, solve a linear system. The cost model of Petit–Kosters–Messeng [10] (read FT; verbatim from the survey) is $P(p, m) + \frac{m!\,p}{(\deg L)^{m-1}}\,T(E, m, L) + (\deg L)^{\omega}$, with $2 < \omega \le 3$; it beats generic methods when $T(E, m, L)$ is below $p^{1/2-1/m}$. The authors of [10] themselves state that asymptotics are open and that all attacks are outperformed by generic discrete logarithm algorithms for practically relevant parameters. The survey also records, at abstract level only, Semaev's binary-curve heuristic [12] (AB; binary curves, prime fields not addressed) and the doubts about the first-fall-degree assumption in characteristic 2 [4, 5] (AB); none gives a prime-field sub-$\sqrt{p}$ algorithm.
**Inventor D (HYP-720e359e, ECDLP-47; $m = 2$ with a Gröbner/enumeration finder).** $S_3$ decomposition over a PKM multiplicative-coset base ($x \in g\mu_d$, $d = 2^n \mid p - 1$): 65/65 runs recovered $k$ (1581 sampled relations verified; sympy Gröbner matches exhaustive search for $p$ up to 17921). Total cost exponent $\sim 1$ (0.99 $[0.92, 1.06]$) against rho $\sim 0.5$; D2a/rho = 11 (16 bits), 94 (20), 633 (24), 2533 (28, post hoc); D2b ($d \sim p^{1/3}$) exponent 1.075; linear algebra $< 0.1\%$. Obstacle: the per-target finder costs $\sim |V|$, a relation costs $\sim 2p/d$ additions, so the total is $\sim p$ independent of $d$; hash and enumeration finders cost at least $p^{1/2-1/(2m)}$ per relation while the accounting needs $T < p^{1/2-1/m}$; Gröbner or resultant elimination collapses to a degree-$|V|$ univariate problem (time $\sim d^{2.1}$ to $d^3$); no subfield of $\mathbb{F}_p$ exists for Weil restriction.
**Inventor E (HYP-3c665e7b, ECDLP-49; $m = 2$ supply).** For fixed $x_3$ the discriminant of $S_3$ in $x_2$ is $16f(x_1)f(x_3)$ with $f = x^3 + ax + b$: the decomposition curve has genus 1 and $j = j(E)$ and is parametrised by the group law itself (affine count $\#E - 2$, brute force on 15 curves, $p = 101$ to 2053). Pulling back to a factor base $x = \varphi(t)$ of degree $e$ gives genus 1 ($t$), 5 ($t^2$), 19 ($t^3$), 9 (generic degree-2 rational), computed exactly; $e \ge 4$ from formulas only. Genus $\le 1$ forces $e = 1$ (the whole of $\mathbb{F}_p$); a base of size $\sqrt{p}$ needs $e \ge \sqrt{p}$, genus $\sim 3p$–$4p$, and the Weil bound is vacuous. Finder-independent supply bound: a random-like base of size $s$ has about $2s^3/(3p)$ closed $S_3$ triples (measured within 2% on four bases), $s$ relations need $s \ge 1.22\sqrt{p}$, so total cost $\ge 1.22\sqrt{p}$ even with a free finder: this ties rho in exponent and cannot beat it. Rational isogenies on prime-order groups are isomorphisms on $\mathbb{F}_p$-points; CM orbits on $j = 0$ give only $\mathbb{Z}[\omega]$ identities with no useful relations. Ledger notes: toy sizes $p \le 2^{16}$; of the prior

literature only abstracts of two papers were seen, and Diem 2011 and PKM were not retrieved at that time (PKM second-hand via inventor D).

**Inventor F (HYP-be407930, ECDLP-52; $m = 3$).** Supply: closed $S_4$ quadruples among $s$ base points number $\sim 8\binom{s}{4}/r \sim s^4/(3r)$, so $s \geq (3r)^{1/3}$; observed/predicted closure 0.991–1.005 on six base families (multiplicative coset, small-$x$, random-$x$, quadratic and cubic images, $gt^3$) at 18–27 bits, so structure on $x$ gives no excess closure. The best finder (a 2+1 split hash with a pair-sum table) solved 24/24 coset instances end to end (16–30 bits; controls 12/12 within 0.91–1.06) with total cost $\sim 1.5\, r^{2/3}$, fitted exponent 0.651 $[0.642, 0.661]$ against 0.5; 8.6× rho at 16 bits and 36× at 30 bits. Per-relation cost is $\sim r^{1/3}$ (37 to 920 additions) against the required $r^{1/6}$. Obstacles: the supply floor holds for any finder; linear algebra $\sim 4s^2$ (a model, not a proved bound) already exceeds $\sqrt{r}$; hash finders cost at least $\min_s(s^2 + r/s) \sim r^{2/3}$; Gröbner degrades to enumeration (sympy: $D = 6$ 29 s, $D = 8$ 216 s, $D \geq 10$ timeout); $x$-structure is not group structure, and group-compatible structure (kernel cosets, endomorphism orbits) is empty or trivial in prime order. The kernel-coset 4-distinct refinement was post hoc.

**Literature survey and PI derivation (HYP-7c98bfbd, ECDLP-58).** The survey (2015–2026) found no surviving claim of a sub-$\sqrt{p}$ algorithm for generic prime-order prime-field curves. It also observed that the supply bounds of inventors E and F count only base-closed relations, whereas PKM-style relations use randomised targets $R = aP + bQ$ with unlimited supply. *PI derivation (unreviewed):* a random target decomposes over a base of size $s$ into $m$ points with probability $\sim s^m/p$ (up to $m!$), about $s$ relations are needed, so total cost $\geq (p/s^{m-1})\, T_{\text{test}} + \text{LA}(s)$. For $m = 2$: $\max(p/s, s) \geq \sqrt{p}$ even with a free finder and linear-time linear algebra (inventor E's conclusion stands). For $m = 3$ with LA $\sim s^2$: $p/s^2 + s^2 \geq \sqrt{p}$ (a tie at best, consistent with inventor F and PKM). For $m \geq 4$: $s = p^{1/(m+1)}$ would give $p^{2/(m+1)} < \sqrt{p}$ only if $T_{\text{test}} < p^{1/2-2/(m+1)}$ ($m = 4$: $< p^{1/10}$ at $s = p^{1/5}$), whereas generic list-merge costs $\sim s^{m/2} = p^{2/5}$. This made the $m \geq 4$ finder the single open index-calculus route.

**Inventor I (HYP-4825a5fb, ECDLP-59; $m = 4$).** Negative: no finder reaches $T_{\text{test}} < p^{1/10}$ (pre-registered rule: $\beta$ with 95% interval upper bound below 0.10), so no pipeline was built. Seven sizes (20–44 bits, $s = 15$ to 457), 3 curves × 20 random + 5 planted targets; all finders decompose 105/105 planted targets (point-verified), and MITM and the resultant tower agree on every random target. Fits $\alpha p^{\beta} + \gamma$: 2+2 hash MITM counted additions $\beta = 0.405$ $[0.37, 0.45]$ (2/5 by construction), wall time 0.475 $[0.44, 0.52]$; resultant tower or $S_3$ chaining (python-flint) 0.585 $[0.55, 0.64]$; 3+1 triple-sum table $T_{\text{test}}$ slope 0.203 but precomputation 0.595 (total above $\sqrt{p}$). Gröbner (Singular 4.4.1 via a wheel in the scratchpad, a recorded protocol deviation): $m = 4$ 4.3 s at $d = 6$ and timeout for $d \geq 8$; $m = 3 \sim d^{5.3}$; Gröbner data are from single curves with no interval. Success per target $\sim s^4/(24p)$ (within 0–60%, converging by 28 bits). Obstacle: chaining finders must handle $s^2/4$ pair sums (a degree-$s^2$ polynomial without closed form on the coset base), so $T_{\text{test}} \geq s^2 = p^{2/5}$; the base is multiplicative in $x$ while the group law is additive in logarithms, so no character-sum shortcut exists without logarithms. Two further statements are derivations only, unreviewed: $j$-sum table splits give best total $p^{m/(2m-1)} > \sqrt{p}$ for every $m$ (analytic), and a Coppersmith-type small-root bound $\sim p^{1/80}$ versus $p^{1/5}$ needed. The ledger says the $p^{m/(2m-1)}$ table bound needs review before any general claim.

**Evidence level.** Experiment at toy sizes ($p \leq 2^{16}$ for E; 16–30 bits for F; 20–44 bits for I) with a Python implementation, plus *unreviewed derivations* (the PI accounting formula, the $p^{m/(2m-1)}$ bound, the $p^{1/80}$ bound, the linear-algebra model). The ledger records index calculus on generic prime-order prime-field curves as *closed for* $m = 2, 3, 4$ *at toy scale*; this is an empirical closure, not a proof of impossibility. No independent review; of the literature only [10] was read at full-text level.

### 8.3 Family 3: ambient groups, base change and isogenies (frontier F, inventor B, frontier I part 2)

**Mechanism.** Move the problem to a larger group (base change $E(\mathbb{F}_p) \subset E(\mathbb{F}_{p^n})$, cover Jacobians, Frey–Kani gluing, generalized Jacobians) or to a different curve (automorphism orbits, isogenies) in the hope of cheaper relations or search.

**Frontier F (HYP-ae6977f0, ECDLP-53).** No ambient group of types (a)–(d) escapes the supply bound. *Universal lemma (heuristic, unreviewed):* for any group $G$ with a computable homomorphism (or cofactor map) onto the $r$-part of $E(\mathbb{F}_p)$, relations in $G$ push down to relations on $\mathbb{Z}/r$; $G$ loses relations by a factor $|G|/r \geq 1$ and can help only through a cheaper finder; the accounting exponent for genus $g = 1$ is $\geq 1/2 + 1/(2(m-1))$ with a generic list-merge finder, and $\geq g$ for $g \geq 2$. (a) Base change: for odd $n$ the base is $V = E(\mathbb{F}_p)$; for even $n$ it lies in $E(\mathbb{F}_p) \times E^{\mathrm{tw}}(\mathbb{F}_p)$ (a direct product); on toy $p = 1009, 2003, 4001$ closed triples were only pure $E$-type or pure twist-type, 0 mixed; Weil restriction gives 1 equation in $n$ unknowns since $R$ is already in the base. (b) Cover Jacobians: $|J| \sim p^g$ forces $s \sim p$ or long relations; exponent $\geq 1$ ($g = 2$), $\geq 4/3$ ($g = 3$). (c) Frey–Kani gluing equals (b) with $g = 2$ ($N = 2$ works over $\mathbb{F}_p$ via a Frobenius 3-cycle on $E[2]$). (d) Generalized Jacobians: the torus is irrelevant to the $r$-part; the $\mathbb{F}^*_{p^k}$ transfer needs a small embedding degree. Ledger caveats: the cover computation uses a product map, not true Jacobian arithmetic; the $(2,2)$-isogeny kernel is ignored; $J$ counts are Poisson-noisy; the cited prior work (Gaudry 2009, Diem 2011, Frey–Kani 1991, Kani 1997) was not retrieved, only recollected, and is therefore not cited here. Open at that point: structured bases with more-than-random closure; a finder for closed $m$-tuples ($m \geq 4$) cheaper than $s^{\lceil m/2 \rceil}$.

**Inventor B (HYP-d6cc9bce, ECDLP-43).** B1: automorphism-orbit rho ($j = 0$, $|\mathrm{Aut}| = 6$; $j = 1728$, $|\mathrm{Aut}| = 4$); B2: 2-isogeny volcano ascent to $j = 0$, then B1. B1: 450/450 runs, $r \leq 2^{36}$; fitted exponents 0.44–0.49 (intervals contain or nearly contain 0.5); pooled rho/orbit ratio 1.45 $[1.14, 1.88]$ ($j = 0$; the pre-registered rule requires the lower bound above 1.15 and it is 1.14) and 2.62 $[2.15, 3.17]$ ($j = 1728$, inflated by a heavy-tailed rho baseline 2–3× above textbook). B2: 150/150; isogeny evaluations $\sim a^{1.86}$ (polylogarithmic); the pre-registered rule printed "asymptotic improvement" but this is a false positive from additive overhead shrinking 60% → 3%; medians indicate exponent 1/2 with constant ∼ 2–2.5. Obstacle: a known technique with constant $\sqrt{|\mathrm{Aut}|}$; B2 applies only to curves isogenous to $j = 0$ (fraction $\sim p^{-0.47}$ $[-0.58, -0.38]$) and is equivalent in kind to Jao–Miller–Venkatesan and Galbraith volcano climbing (named in the ledger; not cited here). Isogeny-pooled, twist-pooled and other curve models were rejected on paper because they preserve $r$ and group structure.

**Frontier I part 2 (HYP-8b156dfd, ECDLP-60; isogeny distance to weak curves).** Order, trace, embedding degree, anomalous and supersingular properties are isogeny-invariant (380 Vélu 2-isogeny steps: order never changed, discrete logarithm transported 380/380), so the distance to a weak curve is 0 or infinite. $j = 0$ or 1728 is reachable only if the CM field is $\mathbb{Q}(\sqrt{-3})$ or $\mathbb{Q}(i)$, a class fraction $\sim 2.5/\sqrt{p}$ ($\beta = -0.49$ $[-0.55, -0.43]$, 32 primes up to $1.07 \cdot 10^6$), with a gain of at most $\sqrt{6}$; every curve in such a class has non-scalar endomorphisms of degree $\geq |d_K|/4$. Ledger caveat: the ascent cost model for degrees $> 2$ is unvalidated.

**Evidence level.** Experiment (toy sizes; for B, $r \leq 2^{36}$) together with one *unreviewed heuristic lemma* (frontier F's universal lemma). No independent review.

### 8.4 Family 4: factor bases with excess group closure (inventor G)

**Mechanism.** Look for an efficiently describable base $V \subset E(\mathbb{F}_p)$ whose sums land in $V$ more often than for a random set (excess closure), which would raise relation supply.

**Cost.** HYP-73c24d12 (ECDLP-55): coordinate-defined sets ($x/y$ windows, joint boxes, bit patterns, low Hamming weight, order statistics, arithmetic progressions in $x$, $xy$ windows) have $m = 3$ closure ratio $1.00 \pm 0.02$ at 14–20 bits. The exception is $m = 2$: $y$-windows and linear-form windows (unions of whole lines) have excess closure growing with $p$ (slope 0.29 $[0.27, 0.32]$ in $\log_2$

per bit; ratio 4.9 to 26 over 14–22 bits at $s = 2.5p^{1/3}$) because the three points of $E$ on a line sum to $O$; but the relations lie on disjoint lines and the relation-matrix rank is only $0.17$–$0.19\,s$, so the exponent stays $1/2$ (a post hoc finding, in a hashed protocol addendum). Images of short vectors ($aG_1 + bG_2$) have ratio 52 to 520 but 99.5–100% trivial relations, target coverage collapses, and membership needs a logarithm. Local search finds high-closure sets (ratio 8–25) that are arithmetic progressions in log space (their description *is* the logarithms).

**Obstacle.** *PI derivation (unreviewed):* a base that is a union of $L$ lines has about $L$ rational points and about $L/6$ relations (lines with 3 rational roots), so rank $\leq\sim s/6$ for any number of pencils, matching the measured $0.17$–$0.19\,s$; more relations per point would need approximate closure under the group law (an approximate subgroup, small-doubling set), which in a prime-order cyclic group means log-space progressions whose membership test needs the logarithms. The residual question, an efficiently testable small-doubling subset of $E(\mathbb{F}_p)$ without logarithms, is recorded as expected to be excluded by sum-product type results for elliptic curves (a literature check not performed).

**Evidence level.** Experiment at 14–22 bits plus an unreviewed derivation. Ledger caveats: the pre-registered prediction 1 failed for $y$-windows, the extreme-value bound was mis-specified ($6\times$ counting), and 22-bit E1 runs were aborted under CPU contention (kept in an `aborted/` directory, unused).

### 8.5 Family 5: lifted relations (inventor A)

**Mechanism.** HYP-23351e75 (ECDLP-42): A1, Farey-height factor-base index calculus with meet-in-the-middle relation search; A2, LLL (Coppersmith-style) decomposition into two Farey points; A3, lift to $E(\mathbb{Q})$ on $y^2 = x^3 - 2$.

**Cost.** A1: 66/66 correct (12–24 bits); cost against rho at 24 bits $42.7\times$ (4-term) and $28.9\times$ (6-term); the 3-term control $\sim 260\times$ at 20 bits; measured exponents 0.82 and 0.71 (post hoc local fits on sizes $\geq 18$ bits) against a rough model 0.667 and 0.60; pre-registered verdict NO-ADVANTAGE. A2: 0 of 124/77/61 existing decompositions found at 16/20/24 bits. A3: planted 50/50, random 0/300 (coverage $10^{-3}$ to $4 \times 10^{-7}$).

**Obstacle.** A1: with $s$ unknown logarithms, hashing needs $\sim \sqrt{ps}$ additions, more than $\sqrt{p}$. A2: rank-one structure is not a lattice condition. A3: random coverage of $10^{-3}$ to $4 \times 10^{-7}$.

**Evidence level.** Experiment, toy sizes, no independent review.

### 8.6 Family 6: precomputation (inventors C and H)

**Mechanism and cost, inventor C (HYP-d913fa61, ECDLP-44).** C1 negation-class rho; C2 preprocessed fixed-walk table; C3 multi-instance shared distinguished-point search. Reference DP rho (31-way) exponent 0.486 $(0.467, 0.506)$ on 20–44-bit toy curves. C1 (frozen) failed: undetected fruitless cycles (period $\geq 6$), $\sim 11.8\sqrt{n}$ at 40 bits, exponent 0.687, 44-bit case censored; post hoc C1b exponent 0.482, pooled gain $\sim 1.13$ (interval $\sim 0.95$–$1.33$), below $\sqrt{2}$, constant-factor rule not met. C2: online exponent 0.344 $(0.317, 0.371)$, preprocessing $\sim n^{0.62}$, $ST^2/n \sim 4$, break-even 15–71 queries. C3: per-instance cost $\sim M^{-0.57}$ $(-0.63, -0.50)$, a $\sqrt{M}$ law, with $n$-exponent unchanged at 0.465 and a gain of $1.7$–$3.1\times$ at $M = 16$.

**Mechanism and cost, inventor H (HYP-bfb7d000, ECDLP-56; fixed-curve precomputation with comparisons and bits).** Negative: no $ST^2/r$ falling with $r$ out of sample (pre-registered rule not met). E1, a sorted $x$-table with interpolation search, is BSGS in disguise (saves probes, not group operations); $ST^2/r$ at $2^{16}$–$2^{32}$: $S = r^{1/3}$ $1.1 \cdot 10^2$–$1.4 \cdot 10^5$, $r^{1/2}$ 44–$3.6 \cdot 10^3$, $r^{2/3}$ 74–141 (16,200 queries verified). E2, greedy comparison decision trees on bits and thresholds of $x(Q)$, $x(Q \pm P)$ (15–4095 nodes, 5 sizes $\times$ 3 curves): held-out accuracy 0.059–0.065 against chance 0.0625, 0/90 cells leak; a planted 2-bit leak is detected (0.25) at every size. E3, curve-tuned walk tables: $ST^2/r = 3.9$ (tuned) against 4.5 (generic) at $2^{32}$, ratio 0.86 $[0.69, 1.06]$; the post hoc 24–32-bit slope $-0.049$ did not replicate on 9 fresh curves ($+0.004$ $[-0.057, 0.064]$).

**Obstacle.** For C: no single-instance exponent below 1/2; C2 lies on the generic $ST^2 \sim N$ trade-off curve and not beyond it; C3 is the known $\sqrt{M}$ law; the ledger classes all of C as known techniques. For H: information about $k$ enters only through exact collisions with advice points (probability $\sim SL^2/r$, a constant ratio); tuning pseudorandom step labels gains only extreme-value constants ($\sim 12\%$); a non-random contracting step function would be needed and none of 7 families worked. The survey records generic-group preprocessing bounds: Corrigan-Gibbs–Kogan [2] (AB) give $ST^2 = \tilde{\Omega}(\varepsilon N)$, tight up to logarithmic factors, and Dinur–Keller–Marmor [3] (AB) show for non-adaptive algorithms with $S$-bit advice that online time $O(\sqrt{N})$ cannot be reduced unless the advice exceeds $\Omega(\sqrt{N})$ bits. These are generic-group statements; the survey says that whether they constrain a non-generic algorithm is not addressed by the sources.
**Evidence level.** Experiment, closed at $r \leq 2^{32}$ for the tested feature and step classes only; *no lower bound exists for a single concrete curve* (ledger). Preprocessing $\sim r^{0.62}$ is reported separately (a different cost measure). Generic-model literature was cited from memory in the H record and was only abstract-level in the survey. No independent review.

### 8.7 Family 7: integer structure, SAT and machine learning (frontiers D and E)

**Frontier D (HYP-93fbd7d0, ECDLP-37).** Mechanisms: (1) structure-aware SAT encodings (x-only Montgomery ladder; native XOR via CryptoMiniSat); (2) the Hensel-section cocycle $g(kP) = k\,g(P) - NS_k$ with Birkhoff sum $S_k$; (3) a lift-variation idea. No cost ratio against rho is recorded. (1) Affine conflict exponents 1.27 (Glucose4) and 1.47 (CaDiCaL); the ladder was unsolved within 120 s at $n = 5$–$7$. (2) The identity was verified on 8 curves ($p \leq 30011$); no low linear complexity, P-recursive or low-degree relation and no $k \bmod m$ dependence in the $p^2/p^3$ digits (the spectral flag was a null-model artefact; symmetry-respecting null $|z| \leq 2.6$, post hoc). All null. Obstacles: structure-aware encodings are worse than the affine encoding; the identity holds but the digits carry no exploitable structure; (3) was killed (4 unknowns against 3 equations; the lift-dependent part is not low-degree). Not reproduced by the PI (long SAT runs); CryptoMiniSat `key_ok` flags were invalid in some rows (indexing bug fixed, 9 instances re-validated by the agent).
**Frontier E (HYP-6da71353, ECDLP-41).** Distinguishers on integer structure of $x(Q)$ at $p = 524309$ (two prime-order curves, seeds $1, 2$): E1 local carry, bit, order and difference correlations of $x(Q)$ versus $x(Q + jP)$; E2 index differences of integer-sorted neighbours; E6 the three-term carry $c_k = [x_{k+1} + x_{k-1} \geq p]$; E4 HNP-lattice; E5 lifting smoothness. No cost ratio or scaling exponent is recorded. E1 null (a 2% planted signal was detected, $z = 28$–$760$); E2 null; E6 fails the pre-registered two-seed rule ($p_T = 0.025$, 0.050 at different $h$; positive control $p_T = 0.005$); E4 and E5 killed by derivation or literature (E5: $\sim 2^{18}$-bit numerators at rank 29 for a 256-bit $p$). Toy size only; absence of signal is not absence of structure; E1 null draws were reduced (60/20) after a timeout, recorded in an amendment. The record also contains two *unreviewed remarks*, E0 and E3 (order and bit access are free in Shoup's model; decision-tree arithmetic length is vacuous without size accounting); the review REV-6c73a098 (decision *revise*, Section 7(iii)) found both hold only in a narrower form, with Theorem DT unaffected and the E0 corollary a restatement of non-genericity.
**Machine learning.** The survey (HYP-7c98bfbd) found as ML "break" only an unvetted single-author preprint (read at AB+SN level, overfitting reported, no replication seen; not cited) and, at abstract level, a gradient-concentration argument limiting learning of the discrete-logarithm parity bit for any architecture [13] (AB). The laboratory's own ML runs are in an earlier round (frontier B) and are not part of this map.
**Evidence level.** Experiment (null searches), toy sizes; E0/E3 reviewed in narrower form only.

### 8.8 Family 8: Problem L (frontier I part 1)

**Mechanism.** HYP-8b156dfd (ECDLP-60). The Las Vegas algorithm of Mahalanobis and Mallick [9] reduces ECDLP to "Problem L", a linear-algebra problem whose solver the authors acknowledge to be the bottleneck. The ledger records the 2017 arXiv paper (v3) as *read in full* by agent RUN-eaefc23d, while the survey table records full-text level with the passages abstract, introduction, Section 3 headings and Section 5.1 read; we state both and the discrepancy is unresolved. A 2026 follow-up [8] was read at abstract level (and, per the ledger, its Section 1.1, Section 5 and Table 2).
**Cost.** The reduction is exactly a zero-sum-subset search: a $3n$-subset of $N = 3n + l$ random multiples of $P$ and $-Q$ summing to $O$ if and only if a singular $l \times l$ kernel-matrix minor exists; verified on 50,488 subsets over 4 toy curves with 0 exceptions. Algorithm 2 has per-pass success $cl^2/r$ ($c \sim 2.5$, flat over $r = 190$ to 29,000) and total cost $\sim 2.4\,r$ group operations; exhaustive Problem L fit $\beta = 0.94$ $[0.82, 1.0]$; generic MITM subset-sum $\alpha \sim 3.5\sqrt{r}$ against a rho control $\sim 3.0\sqrt{r}$; the follow-up paper's own Table 2 reports exponent $\sim 1$. The survey independently notes that the success probability stated in [9], $0.6(\log p)^2/p$ per polynomial-time pass, needs about $p/(\log p)^2$ passes, i.e. exponent 1.
**Obstacle.** The search is a zero-sum subset problem of cost $\sim r$; no sub-$\sqrt{r}$ solver is known to the laboratory.
**Evidence level.** Experiment on toy curves, with these ledger caveats: the MITM and exhaustive solvers use a discrete-logarithm shortcut validated by the equivalence experiment; Algorithm 2's real linear algebra was run only for $n \leq 3$; the rho free-exponent fit is ill-conditioned (a model $\beta = 1/2$ was used). No independent review.

### 8.9 Family 9: transfer of an ETH-refutation technique

**Mechanism.** HYP-32e353ac (ECDLP-46; source "openai/math family 378", external, not peer reviewed, not in the bibliography): transfer of an ETH-refutation technique to ECDLP via a direct CNF (SAT) route, a summation-polynomial decomposition, predicates on $x$ alone, and output bits of $x(kP)$.
**Cost.** Negative on every route, no prototype (decision gate not met). Direct CNF: affine encoding $\sim 48\,\ell^{2.83}$ variables, ladder $\sim 503\,\ell^{2.41}$; the source's parameters (Lemma 4.8, $\beta = 1$) only become non-vacuous at $\log_2 n \sim 10^{10}$; the summation-polynomial route costs $p^{2/(m+1)}$ total even with a free oracle; the $S_3$ test encoding has $\sim 45.5(\log_2 p)^2$ variables; the ANF density of output bits is $\sim 0.5$ up to degree $\ell$.
**Obstacle.** A time bound $2^{Cn/\sqrt{\ln n}}$ beats rho only if $n/\ell < \sqrt{\ln n}/(2C)$, impossible for $n = c\,\ell^a$ with $a > 1$ (extrapolation from $\ell \leq 16$ is crude, but the ledger records the crossover inequality as exact); predicates on $x$ alone are $Q$-independent; there is no sparse ANF form. The parameter thresholds depend on the agent's reading of Sections 3–5 of the source.
**Evidence level.** Unreviewed derivations and model computations without a cost ratio against rho.

**Reading the map.** In families 2, 5 and 8 the measured or derived total cost is at or above rho's (inventor E's supply bound ties rho at best) (ratios stated above); families 3 and 6 report constant-factor effects of known techniques, the fitted exponents 0.44–0.49 (B1), 0.482–0.486 and 0.465 (C) being subject to the methodological note above (inventor B's counted operations include table-construction setup and were fitted as pure power laws, so setup bias is plausible; inventor C excluded one-time table setup from the fitted operations by protocol, but its pure power-law fits leave other fixed overheads unmodelled), and the only recorded exponent below 1/2 is the *online* exponent 0.344 of the preprocessed table C2, whose preprocessing costs $\sim n^{0.62}$ (a different cost measure, on the generic trade-off curve). Families 7 and 9 are null structure searches and model computations without a cost ratio. The ledger outcome *refuted* refers to the

hypothesis that the candidate beats rho or carries a usable signal at the tested sizes. The map is the laboratory's measurement record, not a survey of the literature on ECDLP algorithms, and it makes no novelty claim for the known techniques it re-measured.

## 9 Limitations

- Degree is not time. A lower bound on $\deg g$ for polynomials in one variable $x(kP)$ does not bound the running time of any algorithm, and we draw no bit-complexity consequence.
- The results concern interpolation from $x$-coordinates only, with exact agreement on a range (B, C) or a subset (D); they say nothing about approximate or probabilistic computation or about other coordinates.
- Existence and tightness: interpolating polynomials of degree $\leq N-1$ exist by Lagrange interpolation, so the bounds are tight up to a constant factor; we prove no exact value of the minimal degree (see the conjectural remark in Section 7).
- The ledger's rational-function versions (Theorem A, part (1) of the partial-agreement hypothesis) and the density corollary are not formalized.
- Theorem D's extra hypothesis is removed over $\mathbb{F}_p$ only by an informal Hasse-bound argument.
- The straight-line-program bounds of Section 7(i),(ii) are for restricted, branch-free, single-input models (with and without division); the bit-cost corollary is not formalized. Theorem DT covers equality branching only, charges constants, and is vacuous for arithmetic length once order comparisons are allowed. None of these models contains rho, baby-step giant-step or index calculus.
- The method barrier (iv) is an unreviewed proof attempt, the conjecture C* (v) is unproved, and Theorem E (vi) is a reviewed paper proof that is *not* formalized and was tested only at $N = 3, 4$.
- The obstruction map of Section 8 consists of toy-size experiments and unreviewed derivations without independent review (the only reviewed items are the narrower E0/E3 remarks); its numbers are copied from the ledger and were not recomputed for this report. Fitted exponents there are small-size fits, subject to the methodological note of Family 1. Literature read levels are mostly abstract level; the closure of index calculus at $m = 2, 3, 4$ is empirical (toy scale), not a proof of impossibility.
- No improvement over Pollard rho is claimed.
- The universal $O(n \log n)$ ECDLP target remains open; nothing here improves on any known ECDLP complexity.
- Novelty is unsettled (Section 6); no human has verified the mathematics, the Lean-to-statement correspondence, or the bibliography.

## 10 Conclusion

**No improvement over Pollard rho is claimed.** Every algorithmic route tested in rounds 11–19 (Section 8) either reproduces the generic $\sqrt{r}$ cost, is slower at the tested sizes (index calculus: exponent $\sim 1$ for $m = 2$ with a Gröbner finder, supply floor $1.22\sqrt{p}$ for $m = 2$ even with a free finder, $\sim 1.5\, r^{2/3}$ for $m = 3$, $T_{\text{test}} \geq p^{2/5}$ against the needed $p^{1/10}$ for $m = 4$; lifted relations

28.9×–42.7× rho at 24 bits; Problem L $\sim 2.4r$), gains only constants of known techniques (automorphism orbits, negation, multi-instance batching, preprocessing on the $ST^2 \sim N$ curve), or finds a null signal (integer structure, SAT, comparison trees). Apparent exponents below 1/2 in the ledger are either traced to setup cost (frontier G), online exponents of a preprocessed table (C2), or small-size power-law fits of known $\sqrt{r}$ techniques that were not re-fitted with a setup term (B1, C; see "Reading the map"); this motivates the methodological rule that speedup claims fit $\alpha\sqrt{r} + \gamma$ or use a same-implementation rho control. The contribution of this report is thus (i) a quantified *obstruction map at toy scale*, each family with its mechanism, cost, exact obstacle and evidence level, and (ii) the formal barriers already in version 2: Lean-verified degree bounds for interpolating the discrete logarithm from $x$-coordinates (Theorems B, C, D) and for division-free and rational straight-line programs, and the equality-branching decision-tree barrier (Theorem DT), each with its restricted scope. Nothing here is a lower bound for unrestricted algorithms, an upper bound better than known ones, or a statement for large parameters; the toy-scale evidence is not independently reviewed, and the universal $O(n \log n)$ ECDLP target remains open. The remaining levers named by the ledger are non-generic structure (CM or automorphism families, isogeny distance to structured curves), a finder for $m \geq 4$ decompositions, and a review of the E0/E3 remarks and of the $p^{m/(2m-1)}$ bound.

**Provenance note.** This report was produced by an AI research laboratory (this draft and its revisions 1, 2 and 3 by worker agents reporting model identity Sonnet 5.5, `claude-sonnet-5-5`, not verifiable from runtime metadata) and reviewed by AI agents; no human verification has yet taken place. It is an internal technical report and not a submission.

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